

Report: Cardio Risk Prediction

1. Exploratory Data Analysis (EDA)

Your report should present a clear and thorough EDA of the training data, including data cleaning steps, univariate and bivariate analyses, numerical feature distributions and outliers, correlation analysis, class balance checks, target likelihood analysis for categorical variables, and relationships between numerical features and the target.

Use appropriate visualisations with meaningful interpretations that link back to the business objective.

1.1 Data Cleaning

After loading the dataset, I first checked the data and statistical summary to understand the values. I noticed that some columns had values that did not look realistic, so I cleaned them before continuing with the analysis.

- **Blood pressure (ap_hi, ap_lo):** I removed very low and very high values that appeared unrealistic. I also removed rows where diastolic blood pressure was greater than or equal to systolic blood pressure.
- **Height:** I removed values below 100 cm and above 250 cm.
- **Weight:** I removed values below 30 kg and above 200 kg.
- **Age:** Age was given in days, so I converted it into years to make it easier to understand.
- **Height conversion:** Height was converted from centimetres to metres because I later used it for BMI calculation.
- **Categorical columns:** Columns such as gender, cholesterol, gluc, smoke, alco, active and cardio were treated as categorical variables.

After cleaning, 68,646 records remained from the original 70,000 records. This still gave enough data to continue the analysis and model training.

1.2 Univariate Analysis — Numerical Features

Histograms were plotted for all five numerical features: age, height, weight, systolic blood pressure (ap_hi), and diastolic blood pressure (ap_lo). Key observations:

- Age is distributed mostly between 40 and 65 years.
- Height and weight show fairly regular distributions after the data cleaning.
- Blood pressure (ap_hi and ap_lo) has noticeable peaks around some commonly occurring values.

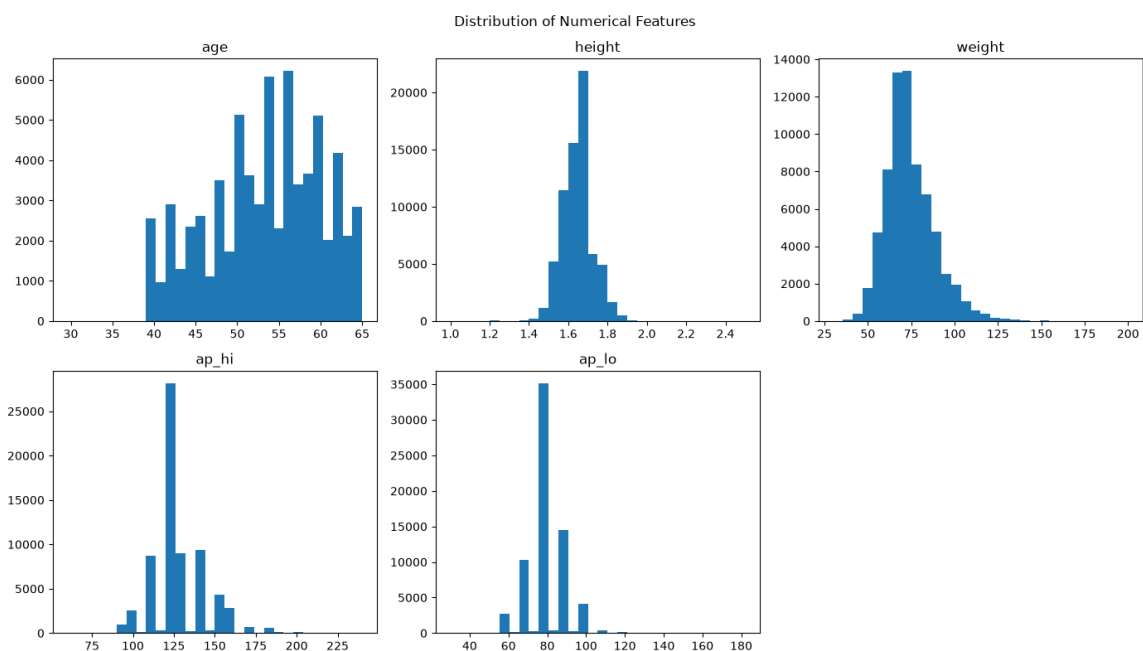


Figure 1: Distribution of Numerical Features

1.3 Univariate Analysis — Categorical Features

Bar charts were plotted for the six categorical features. Key observations:

- Most patients are female. The gender split is approximately 65% female, 35% male.
- The majority of patients have normal cholesterol (category 0). Above-normal and well-above-normal groups are smaller.
- Glucose follows a similar pattern with the vast majority in the normal category.
- Smoking and alcohol intake are reported by less than 10% of patients in each case.
- Most patients (around 80%) are physically active.

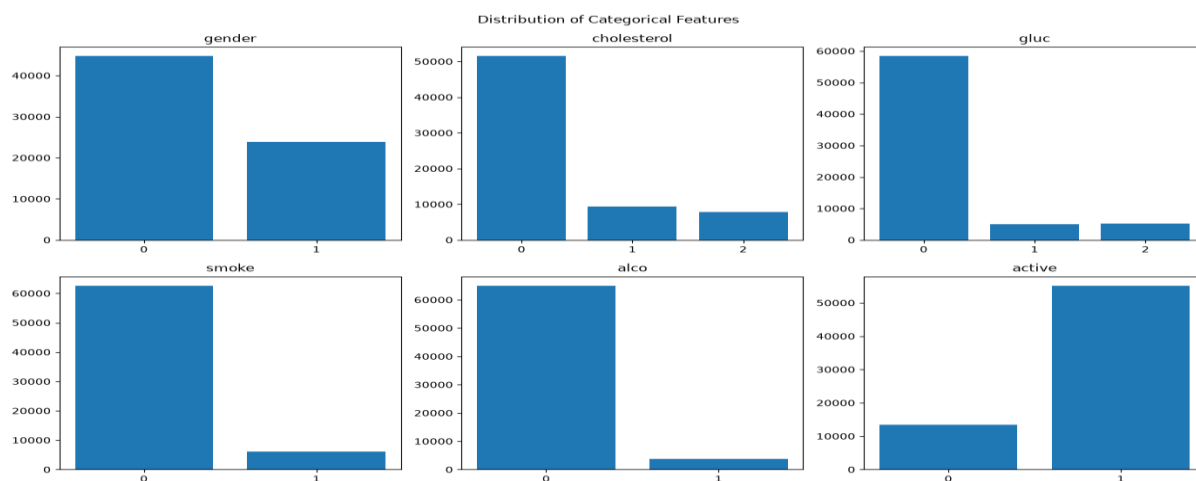


Figure 2: Distribution of Categorical Features

1.4 Target Class Distribution

The target variable (cardio) is well-balanced: 34,684 patients (50.5%) have no cardiovascular disease and 33,962 patients (49.5%) have cardiovascular disease. The two classes are almost equally distributed, so I did not apply any additional resampling technique for class imbalance.

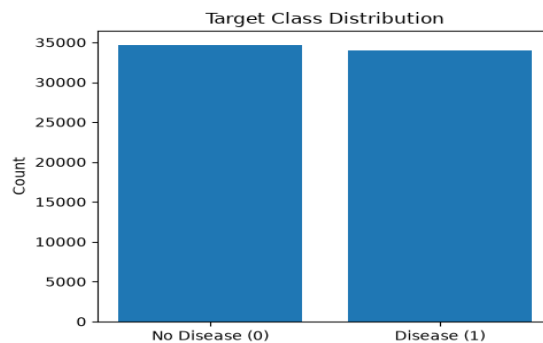


Figure 3: Target Class Distribution (cardio = 0 vs 1)

1.5 Correlation Analysis

The correlation heatmap of numerical features shows:

- ap_hi and ap_lo have a positive correlation of around 0.73, which is expected because both represent blood pressure measurements.
- Age shows some positive correlation with the blood pressure variables.
- Height and weight do not show a strong correlation with most of the other numerical features.
- At this stage, I did not observe any other very high correlation between the numerical features.

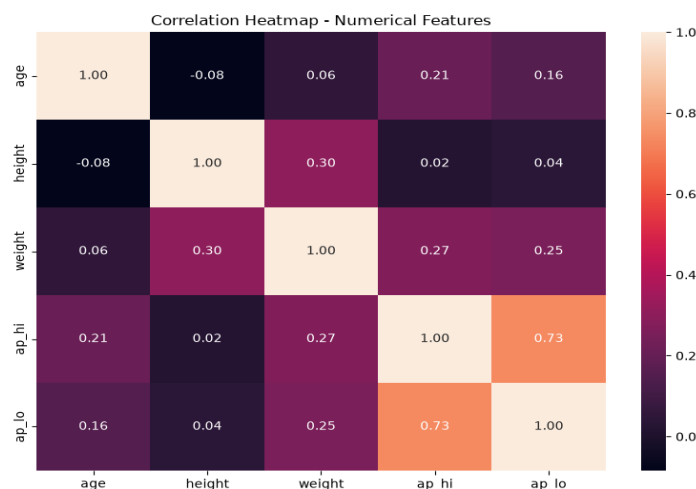


Figure 4: Correlation Heatmap of Numerical Features

1.6 Bivariate Analysis — Categorical Features vs Target

For each categorical feature, the proportion of patients with cardio = 1 was calculated for each category.

- **Cholesterol:** This showed the clearest difference among the categorical features. Patients with above-normal cholesterol had a higher cardiovascular disease rate, around 62%, compared with around 46% for patients with normal cholesterol.
- **Glucose:** A similar pattern can be seen for glucose, although the difference is smaller than for cholesterol.
- **Gender:** The disease rate shows only a small difference between the gender categories.
- **Smoking and Alcohol:** I expected these variables to show a clearer difference, but in this dataset the disease rates are quite similar between the categories.
- **Physical Activity:** Active patients show a slightly lower cardiovascular disease rate in this dataset.

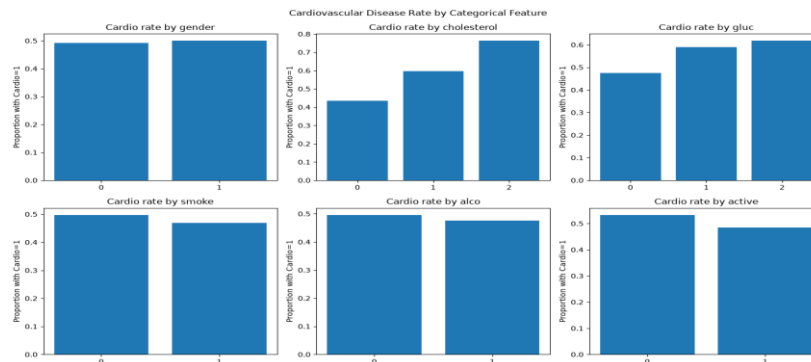


Figure 5: Cardiovascular Disease Rate by Categorical Feature

1.7 Bivariate Analysis — Numerical Features vs Target

Overlapping histograms were plotted for each numerical feature split by the target variable (cardio = 0 vs 1).

- **Age:** From the graph, I observed that patients with cardio = 1 are generally older compared to patients with cardio = 0. This suggests that age could be an important feature for the model.
- **Systolic Blood Pressure (ap_hi):** The cardio = 1 group shows higher systolic blood pressure values overall. This suggests that ap_hi could be an important feature for prediction.
- **Diastolic Blood Pressure (ap_lo):** A similar pattern can be seen for ap_lo, although the difference between the two groups is less pronounced than for ap_hi.
- **Weight:** Disease patients tend to weigh more on average, though the distributions overlap considerably.

- **Height:** I could not see much difference between the two groups for height, so it may not contribute as much as some of the other features.

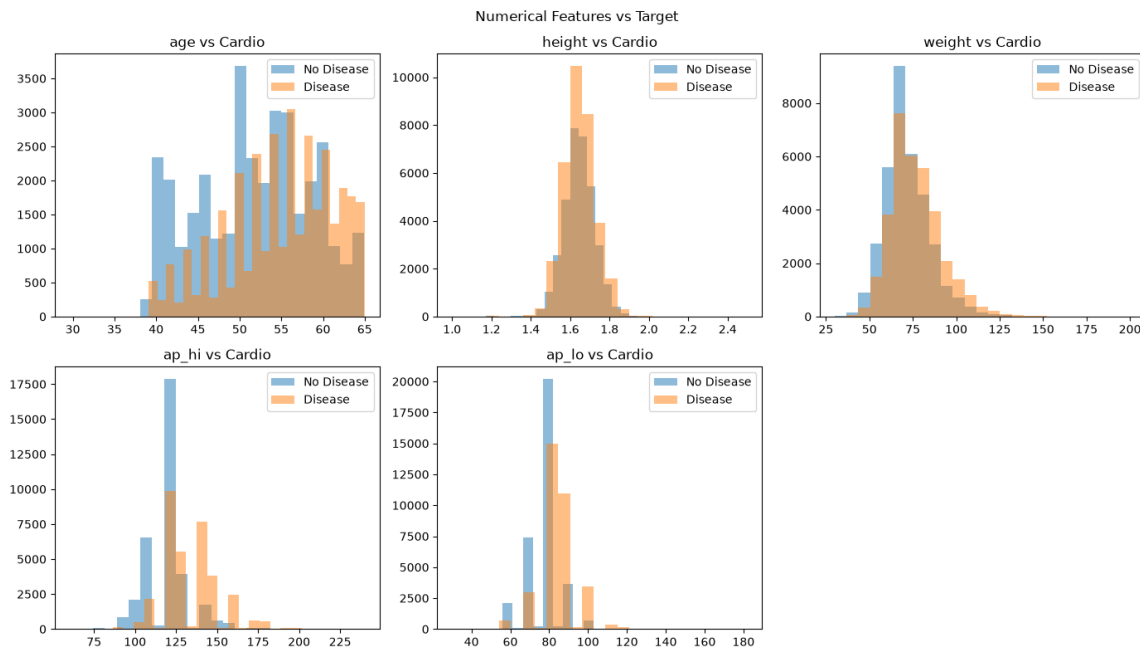


Figure 6: Numerical Features vs Target (overlapping histograms)

2. Model Building and Rationale

Your report should clearly describe all candidate models, the rationale for selecting them, the feature selection or engineering process, the choice of cutoffs, and justification for hyperparameters.

2.1 Feature Engineering

Before training the models, I created and modified some features.

- **BMI:** I created BMI using $\text{weight} / \text{height}^2$. Since BMI already uses height and weight, I checked its correlation with these variables. BMI had a high correlation with weight, so I decided to remove height and weight and use BMI instead.

The correlation heatmap after adding BMI is shown below:

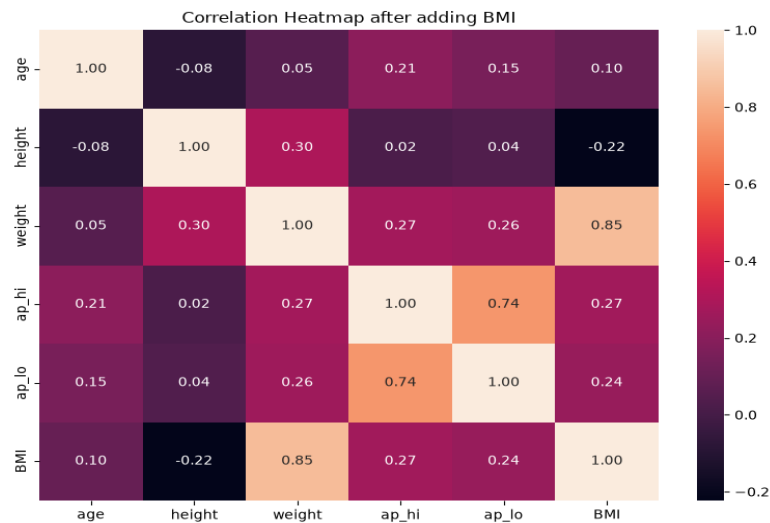


Figure 7: Correlation Heatmap after adding BMI — shows high correlation between BMI and weight

- **Cholesterol and glucose:** I combined the above-normal and well-above-normal categories because their behaviour was similar during the EDA. This also made the categories simpler.
- **Categorical variables:** I converted the categorical columns into dummy variables so that they could be used by the machine learning models.
- **Scaling:** I applied MinMaxScaler to age, ap_hi, ap_lo and BMI. I fitted the scaler using only the training data and then used the same scaler on the test data. I did this so that information from the test data would not influence model training.

2.2 Model 1 — Linear SVM

Rationale: I selected SVM as one of the models because this is a binary classification problem. I started with a linear kernel to establish a baseline and then compared it with other models. I enabled probability estimates because I also wanted to study how changing the probability cutoff affects the prediction results.

For the cutoff analysis:

- I compared `predict_proba()` and `predict()` using the default cutoff of 0.5 and observed similar results.
- I plotted the ROC curve and calculated the AUC to evaluate the model across different thresholds.
- I tested cutoffs between 0.1 and 0.9 and compared sensitivity, specificity and accuracy.
- From this analysis, 0.44 provided a useful balance between sensitivity and specificity. Since missing a cardio-positive case is important in this problem, I also considered recall while comparing the cutoffs.

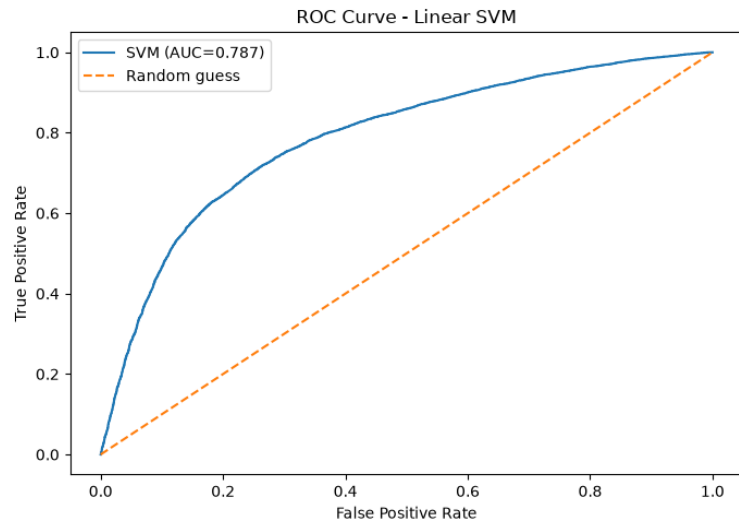


Figure 8: ROC Curve — Linear SVM (AUC = 0.788)

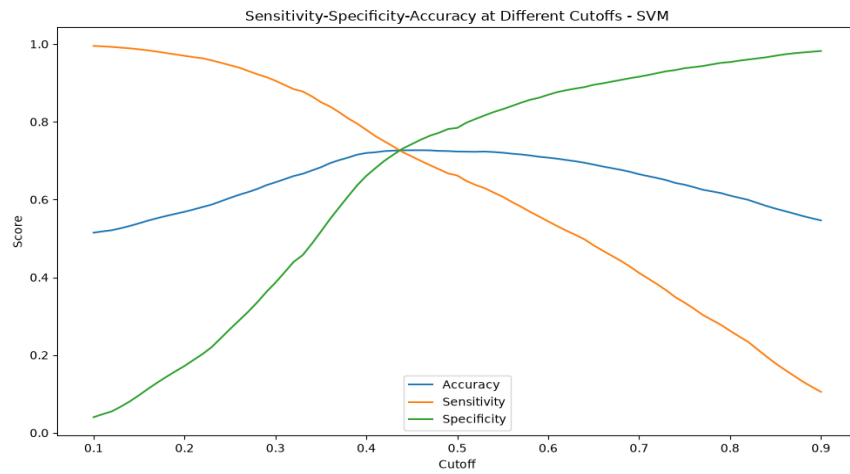


Figure 9: Sensitivity, Specificity and Accuracy at Different Cutoffs — SVM (selected cutoff: 0.44)

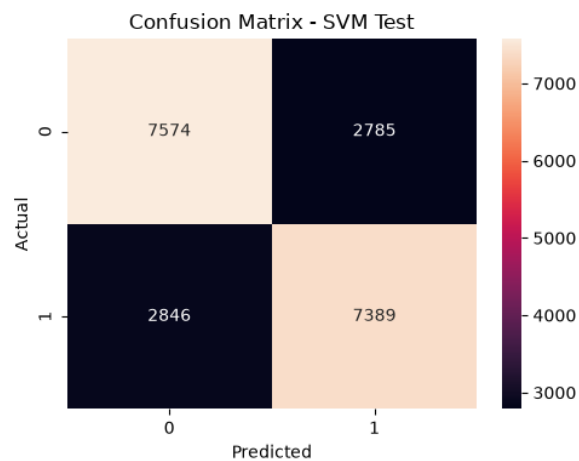


Figure 10: Confusion Matrix — Linear SVM on Test Set

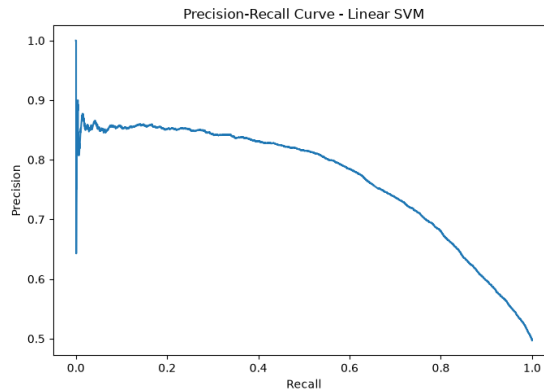


Figure 11: Precision-Recall Curve — Linear SVM

2.3 Model 2 — Decision Tree

Rationale: I selected a Decision Tree because it provides a different approach from the linear SVM and can capture non-linear relationships in the data. It is also relatively easy to understand because its decisions can be represented as rules. Since Decision Trees can overfit the training data, I compared its training and test performance before and after tuning.

Default Decision Tree results:

- **Training Accuracy:** 99.58%
- **Test Accuracy:** 62.58%

The large difference between training and test accuracy indicates that the default Decision Tree is overfitting the training data.

I applied the same ROC and cutoff analysis to the Decision Tree. The default tree gave an AUC of 0.627, which was lower than the SVM result.

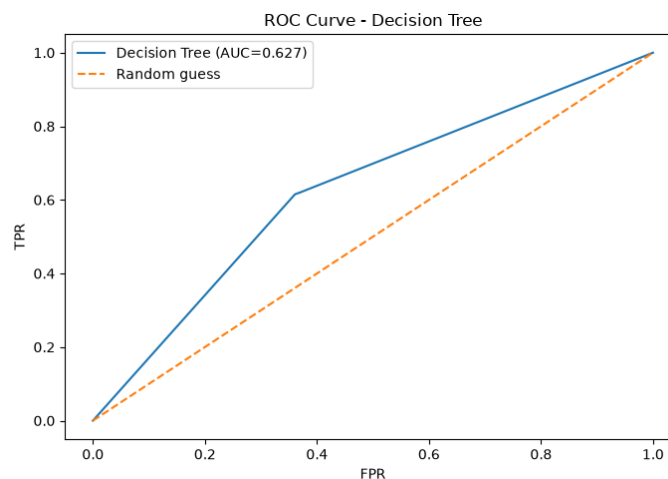


Figure 12: ROC Curve — Default Decision Tree (AUC = 0.627)

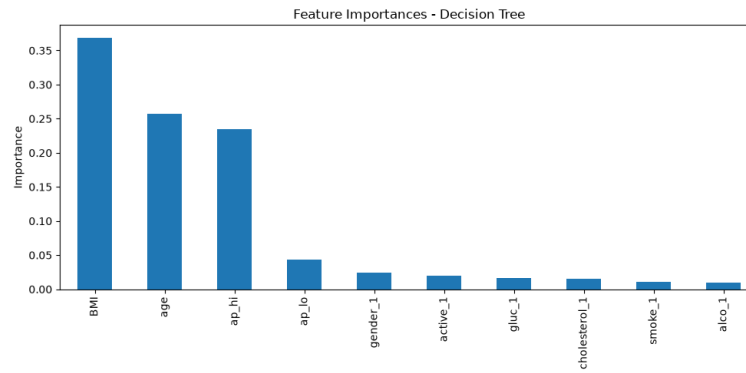


Figure 13: Feature Importances — Decision Tree (*ap_hi*, *age*, and *ap_lo* are most important)

Hyperparameter Tuning (GridSearchCV):

- 5-fold cross-validation was used with `roc_auc` as the scoring metric.
- **Grid searched over:** `max_depth` [3,5,7,10], `min_samples_split` [2,10,20], `min_samples_leaf` [1,5,10], `criterion` [gini, entropy], `max_features` [sqrt, log2].
- **Best parameters found:** `criterion=entropy`, `max_depth=10`, `max_features=sqrt`, `min_samples_leaf=5`, `min_samples_split=2`.
- **Best CV AUC:** 0.779. The tuned model performed better than the default Decision Tree.

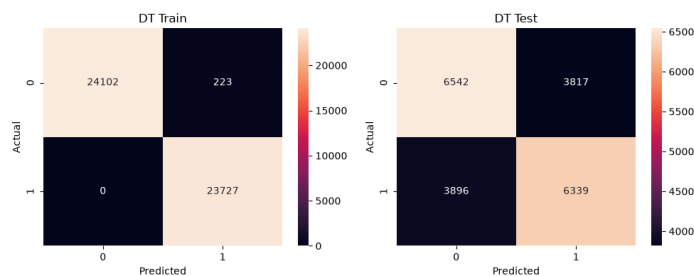


Figure 14: Confusion Matrices — Default Decision Tree (Train vs Test at optimal cutoff)

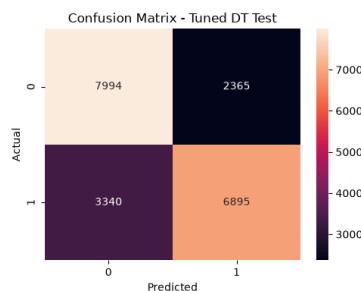


Figure 15: Confusion Matrix — Tuned Decision Tree on Test Set

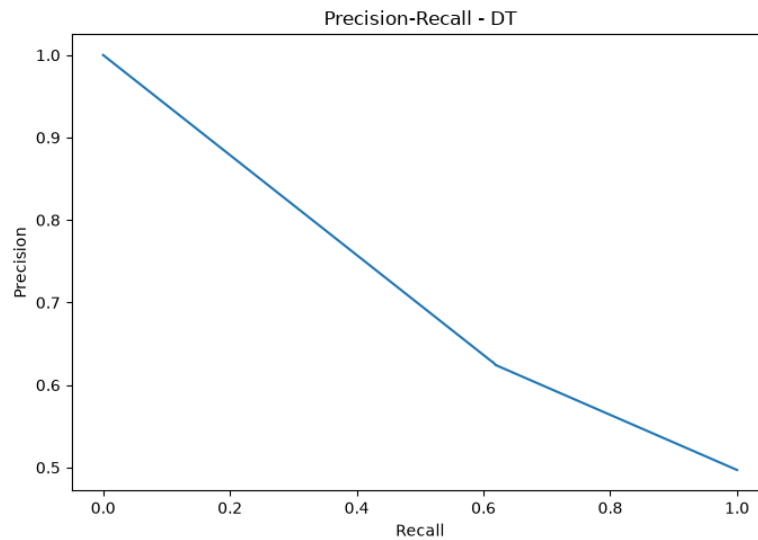


Figure 16: Precision-Recall Curve — Decision Tree

2.4 Model 3 — Random Forest

Rationale: I selected Random Forest after testing the Decision Tree because it combines predictions from multiple decision trees. I wanted to check whether this approach could reduce the overfitting observed in the single Decision Tree and provide better test performance.

- **Default Random Forest (100 trees):** Train Accuracy 99.57%, Test Accuracy 69.83%. The difference between these values shows that the default Random Forest was also overfitting, although its test accuracy was better than the default Decision Tree.
- **GridSearchCV tuning over:** `n_estimators` [100,200], `max_depth` [5,10,None], `min_samples_split` [2,10], `min_samples_leaf` [1,5].
- **Best parameters:** `max_depth=10`, `min_samples_leaf=5`, `min_samples_split=2`, `n_estimators=100`.
- **Best CV AUC:** 0.797, which was the highest cross-validation AUC among the models tested.

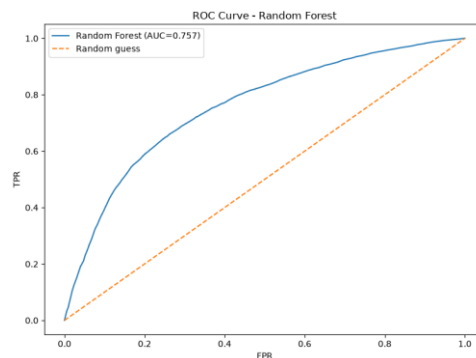


Figure 17: ROC Curve — Random Forest (AUC = 0.757 default, 0.798 tuned)

2.5 Model Performance Summary

The table below summarises the performance of all baseline and tuned models on the test set:

Model	Train Acc	Test Acc	Train Recall	Test Recall	Test AUC
Linear SVM (baseline, cutoff=0.44)	72.33%	72.66%	72.44%	72.19%	0.788
Tuned SVM (RBF, C=10)	72.54%	72.62%	67.45%	67.00%	0.789
Default Decision Tree	99.58%	62.58%	~100%	~62%	0.627
Tuned Decision Tree	73.50%	72.30%	69.16%	67.37%	0.787
Default Random Forest	99.57%	69.83%	99.43%	69.08%	0.757
Tuned Random Forest	74.77%	73.11%	70.38%	67.87%	0.798

Table 1: Model Performance Comparison on Test Set

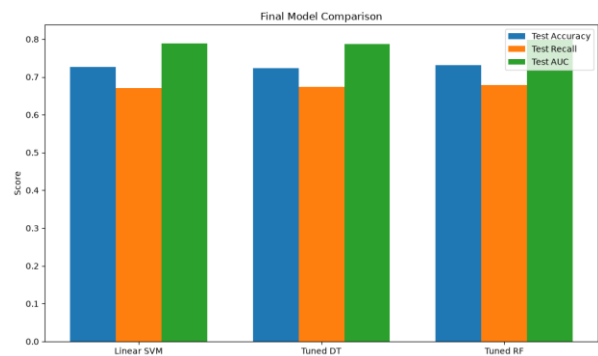


Figure 18: Final Model Comparison — Test Accuracy, Recall, and AUC

3. Model Cards

Your report should include complete model cards for all candidate models, covering overview, intended use, data and features used, model configuration, performance results, and limitations or considerations.

Model Card 1: Linear SVM (Tuned)

Field	Details
Model Name	Linear / RBF SVM — Support Vector Classifier
Model Overview	A Support Vector Machine used for cardiovascular disease classification. I first tested a linear SVM as the baseline and then included both linear and RBF kernels during GridSearchCV. The RBF kernel with C=10 gave the best cross-validation result.
Intended Use	Binary classification to predict whether a record belongs to cardio = 0 or cardio = 1 using the available dataset features.
Primary Task	Supervised binary classification
Data and Features	age, ap_hi, ap_lo, BMI (engineered), gender_1, cholesterol_1, gluc_1, smoke_1, alco_1, active_1 — all scaled with MinMaxScaler.
Engineered Features	BMI = weight / height ² . Cholesterol and gluc categories 1 and 2 merged into 1. Weight and height dropped after BMI creation.
Algorithm	SVC (scikit-learn), RBF kernel, C=10, gamma=auto, probability=True
Hyperparameter Tuning	GridSearchCV, 3-fold CV, scoring=roc_auc. Grid: C=[0.1,1,10], kernel=[linear,rbf], gamma=[scale,auto]
Best Parameters	kernel=rbf, C=10, gamma=auto
Best CV AUC	0.789
Train Accuracy	72.54%
Test Accuracy	72.62%
Train Recall	67.45%
Test Recall	67.00%
Test AUC	0.789
Selected Cutoff	0.44 (selected from sensitivity-specificity analysis)
Strengths	Training and test results are close, indicating limited overfitting. It also gave competitive accuracy and AUC compared with the tree-based models.
Weaknesses	The test recall is around 67%, so the model still misses some cardio-positive cases. The RBF model is also less straightforward to interpret than a single Decision Tree.
Limitations	Performance is based on the features available in this dataset. The model would need to be tested on other data before making assumptions about how well it would perform on a different population.

Model Card 2: Decision Tree Classifier (Tuned)

Field	Details
Model Name	Decision Tree Classifier (Tuned via GridSearchCV)

Model Overview	A Decision Tree model used for cardiovascular disease classification. The default model showed a large difference between training and test accuracy, indicating overfitting. I therefore used GridSearchCV to tune the model and compared the results again.
Intended Use	Binary classification to predict cardio = 0 or cardio = 1. I also used this model to understand the effect of overfitting and hyperparameter tuning on a Decision Tree.
Primary Task	Supervised binary classification
Data and Features	age, ap_hi, ap_lo, BMI (engineered), gender_1, cholesterol_1, gluc_1, smoke_1, alco_1, active_1 — all scaled with MinMaxScaler.
Engineered Features	BMI = weight / height ² . Cholesterol and gluc categories 1 and 2 merged into 1. Weight and height dropped after BMI creation.
Algorithm	DecisionTreeClassifier (scikit-learn)
Hyperparameter Tuning	GridSearchCV, 5-fold CV, scoring=roc_auc. Grid: max_depth=[3,5,7,10], min_samples_split=[2,10,20], min_samples_leaf=[1,5,10], criterion=[gini,entropy], max_features=[sqrt,log2]
Best Parameters	criterion=entropy, max_depth=10, max_features=sqrt, min_samples_leaf=5, min_samples_split=2
Best CV AUC	0.779
Train Accuracy (default)	99.58% ← severe overfitting
Test Accuracy (default)	62.58%
Train Accuracy (tuned)	73.50%
Test Accuracy (tuned)	72.30%
Train Recall (tuned)	69.16%
Test Recall (tuned)	67.37%
Test AUC (tuned)	0.787
Selected Cutoff	0.44 (selected from sensitivity-specificity analysis)
Top Features (by importance)	ap_hi (systolic BP) is the most important feature, followed by age and ap_lo (diastolic BP). BMI is also among the top predictors.
Strengths	The model is comparatively easy to interpret and visualise. After hyperparameter tuning, the difference between training and test accuracy reduced significantly.
Weaknesses	The default model overfit the training data considerably. Even after tuning, its test AUC was slightly lower than the tuned Random Forest.
Limitations	Decision Trees can be sensitive to their parameter settings and changes in the training data. The tuned model also has a test recall of around 67%, so some cardio-positive cases are still incorrectly predicted.

Model Card 3: Random Forest Classifier (Tuned)

Field	Details
Model Name	Random Forest Classifier (Tuned via GridSearchCV)
Model Overview	A Random Forest model that combines multiple Decision Trees. I tested this model to see whether using multiple trees could reduce the overfitting observed in the single Decision Tree. GridSearchCV was then used to find a better combination of parameters.
Intended Use	Binary classification to predict cardio = 0 or cardio = 1. Among the models tested in this assignment, the tuned Random Forest gave the best overall test accuracy and AUC.
Primary Task	Supervised binary classification
Data and Features	age, ap_hi, ap_lo, BMI (engineered), gender_1, cholesterol_1, gluc_1, smoke_1, alco_1, active_1 — all scaled with MinMaxScaler.
Engineered Features	BMI = weight / height ² . Cholesterol and gluc categories 1 and 2 merged into 1. Weight and height dropped after BMI creation.
Algorithm	RandomForestClassifier (scikit-learn)
Hyperparameter Tuning	GridSearchCV, 3-fold CV, scoring=roc_auc. Grid: n_estimators=[100,200], max_depth=[5,10,None], min_samples_split=[2,10], min_samples_leaf=[1,5]
Best Parameters	max_depth=10, min_samples_leaf=5, min_samples_split=2, n_estimators=100
Best CV AUC	0.797
Train Accuracy (default)	99.57%
Test Accuracy (default)	69.83%
Train Accuracy (tuned)	74.77%
Test Accuracy (tuned)	73.11%
Train Recall (tuned)	70.38%
Test Recall (tuned)	67.87%
Test AUC (tuned)	0.798
Strengths	It gave the best accuracy and AUC among the models I tested. After tuning, the difference between training and test accuracy was also much smaller.
Weaknesses	Random Forest is more difficult to understand compared with a single Decision Tree because it contains many trees.
Limitations	The model still does not identify every cardio-positive patient correctly. Also, the results are based only on the features available in this dataset, so adding more health-related features may improve the prediction.

4. Final Conclusion

Your report should summarise key insights from all stages: important findings from EDA, baseline vs tuned model performance, overfitting issues and how they were resolved, data sufficiency, suitability of linear vs non-linear models, the reasoning behind the final model choice, and the final outcomes.

4.1 Key EDA Insights

The EDA helped me identify which features appeared to have a stronger relationship with the target. Age and blood pressure showed noticeable differences between the cardio = 0 and cardio = 1 groups. Cholesterol also showed a clear difference among the categorical variables.

Smoking and alcohol did not show much difference between the two target groups in this dataset, which was different from what I initially expected. The target classes were also almost equally distributed, so I did not apply any class-balancing technique.

4.2 Feature Engineering Impact

The main feature I created was BMI using height and weight. After adding BMI, I checked the correlations again and found a high correlation between BMI and weight. I therefore removed the original height and weight features and used BMI for model building.

I also simplified the cholesterol and glucose categories by combining their above-normal categories based on the patterns observed during EDA. The remaining categorical features were converted into dummy variables, and the numerical features were scaled before model training.

4.3 Overfitting — Problem and Resolution

Overfitting was the dominant challenge for tree-based models:

Model	Train Accuracy	Test Accuracy	Gap
Default Decision Tree	99.58%	62.58%	37.0%
Tuned Decision Tree	73.50%	72.30%	1.2%
Default Random Forest	99.57%	69.83%	29.7%
Tuned Random Forest	74.77%	73.11%	1.7%
Linear SVM	72.33%	72.66%	~0%

Table 2: Overfitting Analysis — Train vs Test Accuracy

The results helped me see the effect of overfitting clearly. Both default tree-based models had very high training accuracy but much lower test accuracy. After tuning the parameters using GridSearchCV, the gap between training and test accuracy reduced considerably.

4.4 Data Sufficiency

After cleaning, the dataset contained 68,646 records, which provided a good amount of data for training and testing the models in this assignment. The two target classes were also almost equally distributed.

The tuned models produced test accuracy around 72% to 73%. Since the results remained fairly close across different algorithms, it appears that changing the model alone may not produce a major improvement. As a next step, I would try adding other relevant features to see whether they improve the model results.

4.5 Linear vs Non-linear Models

The Linear SVM gave 72.66% test accuracy and an AUC of 0.788. The tuned Decision Tree produced similar results, while the tuned Random Forest performed slightly better with 73.11% test accuracy and an AUC of 0.798.

From these results, I observed that the difference between the linear and non-linear models was relatively small. The Random Forest gave some improvement, but using a more complex model did not produce a large increase in overall performance.

4.6 Final Model Selection

Based on the models tested in this assignment, I selected the tuned Random Forest as the final model. It achieved the highest test accuracy of 73.11% and the highest test AUC of 0.798. After tuning, the difference between its training accuracy (74.77%) and test accuracy (73.11%) was also much smaller than with the default Random Forest.

The improvement over the SVM and tuned Decision Tree is not very large, so I would not conclude that Random Forest is significantly better in every situation. However, based on the results obtained from this dataset, it gave the best overall performance among the models I tested.

4.7 Learning from the Assignment

This assignment helped me understand that getting high training accuracy does not always mean that a model is good. The Decision Tree was a good example because it gave almost 100% training accuracy but performed much worse on the test data.

I also learned how EDA helps in understanding which features may be useful before building a model. Comparing the models also showed me that using a more complex algorithm does not automatically give much better results. In this case, tuning had a noticeable effect on the tree-based models, especially in reducing overfitting.

Finally, I learned the importance of using different evaluation measures such as accuracy, recall and AUC instead of depending only on accuracy.